KNEE OSTEOARTHRITIS DETECTION AND SEVERITY GRADING WITH SHAP AND LIME INTERPRETABILITY

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ABSTARCT

Knee osteoarthritis causes pain and functional impairment. This study leverages X-ray imaging and deep learning to classify OA severity using the Kellgren-Lawrence (KL) grading system. Efforts focus on data preparation, augmentation, and visualization for optimal model training, with current progress in dataset management and upcoming phases on training and evaluation.

INTRODUCTION

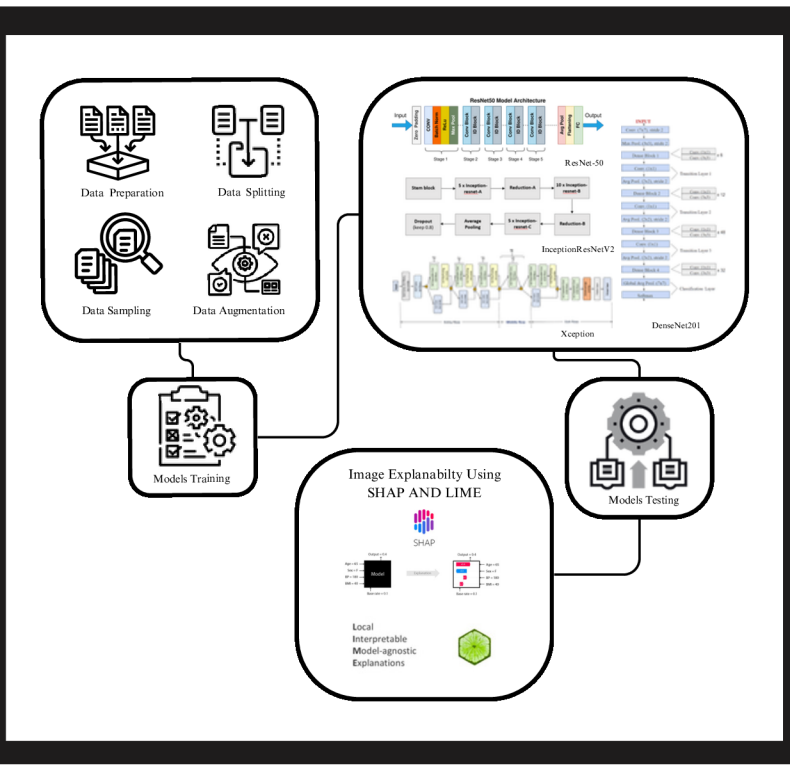
Knee osteoarthritis (OA) is a common degenerative joint disorder causing pain and reduced mobility. Traditional diagnostic methods, like X-ray evaluation with the Kellgren-Lawrence (KL) grading system, are subjective and time-intensive. This study uses AI and CNNs to automate OA severity classification, addressing challenges like dataset imbalances to improve accuracy. Potential applications include mobile apps, telemedicine, and wearable technologies, promoting efficient healthcare.

LITERATURE REVIEW

- Sozan Mohammed Ahmed et al. (2023): Applied machine learning for knee OA detection.
- Tayyaba Tariq et al. (2023): Enhanced knee OA classification using advanced techniques.
- Abdul Sami Mohammed et al. (2023): Improved knee OA grading with machine learning models.
- Fathima Nourin. K (2023): Developed feature-based classification methods for knee OA.
- Ravindra D. Kale et al. (2024): Recent contribution on novel knee OA severity classification methods.
- María Fernanda Rodríguez R. (2023): Ongoing knee OA research shared on GitHub.

METHODOLOGY

This study developed a workflow for classifying knee osteoarthritis severity using X-ray images from Kaggle and the Osteoarthritis Initiative, based on the Kellgren-Lawrence (KL) grading system. The dataset was split into training (80%) and testing (20%) subsets, with data augmentation and class weights to address imbalance. Four pre-trained models (ResNet-50, InceptionResNetV2, Xception, DenseNet201) were fine-tuned for multi-class classification. Model performance was evaluated using accuracy, precision, recall, and F1-score, with SHAP and LIME for interpretability, enabling reliable OA severity classification.



DISCUSSION

This project highlights the potential of machine learning models in healthcare, specifically for knee osteoarthritis (KOA) detection and grading. The deep learning model achieved high accuracy in classifying severity levels from medical images. However, interpretability was a key focus, with SHAP and LIME ensuring that the model's decisions were understandable. SHAP provided global insights into important features, while LIME offered case-specific explanations, enhancing transparency. By addressing the challenge of bridging high performance with human interpretability, the project demonstrates the importance of combining accuracy with explainability for reliable AI in medical applications.

RESULTS

The knee osteoarthritis severity classification models were evaluated, with Xception emerging as the top performer, achieving the highest validation accuracy of 86.62% and balanced accuracy of 90.17%, alongside a test accuracy of 83.71% and macro F1 score of 0.87. Weighted categorical cross-entropy effectively addressed class imbalance, particularly improving recall for severe classes. SHAP and LIME were applied to Xception for interpretability, revealing key features and image region contributions. While Xception outperformed other models, DenseNet201 and ResNet-50 showed strengths in handling severe classes and computational efficiency. Future work may explore ensemble methods to leverage these complementary strengths.

Summary Metrics (Xception):

- Class 0 F1 = 0.84
- Class 1 F1 = 0.72
- Class 2 F1 = 0.86
- Class 3 F1 = 0.96
- Class 4 F1 = 0.99

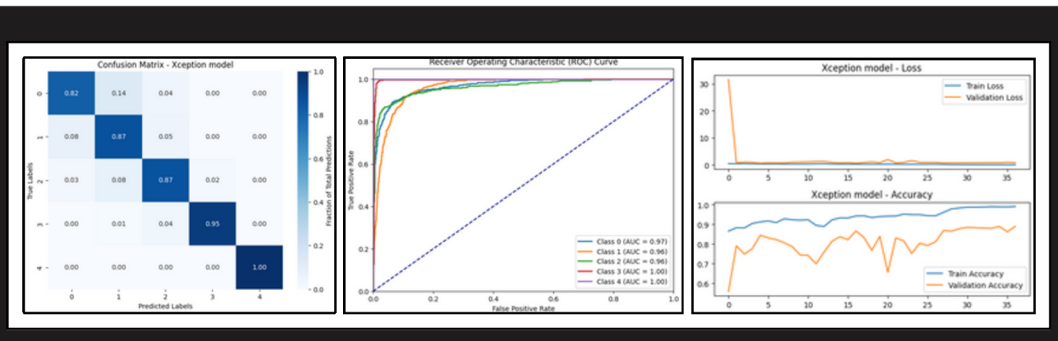


TABLE VI: Summary of Classification Reports for All Models

Model	Class	Precision	Recall	F1-Score	Support	Accuracy
ResNet-50	0	0.83	0.68	0.75	1000	76.78%
	1	0.41	0.59	0.48	530	
	2	0.74	0.69	0.71	757	
	3	0.88	0.89	0.88	380	
	4	0.94	1.00	0.97	83	
InceptionResNetV2	0	0.93	0.71	0.80	1000	84.33%
	1	0.52	0.91	0.66	530	
	2	0.88	0.75	0.81	757	
	3	0.95	0.86	0.90	380	
	4	1.00	1.00	1.00	83	
Xception	0	0.92	0.74	0.82	1000	90.17%
	1	0.67	0.89	0.76	530	
	2	0.85	0.80	0.83	757	
	3	0.90	0.84	0.87	380	
	4	0.98	1.00	0.99	83	
DenseNet201	0	0.77	0.85	0.81	1000	80.42%
	1	0.52	0.49	0.51	530	
	2	0.84	0.69	0.76	757	
	3	0.85	0.99	0.91	380	
	4	0.99	1.00	0.99	83	

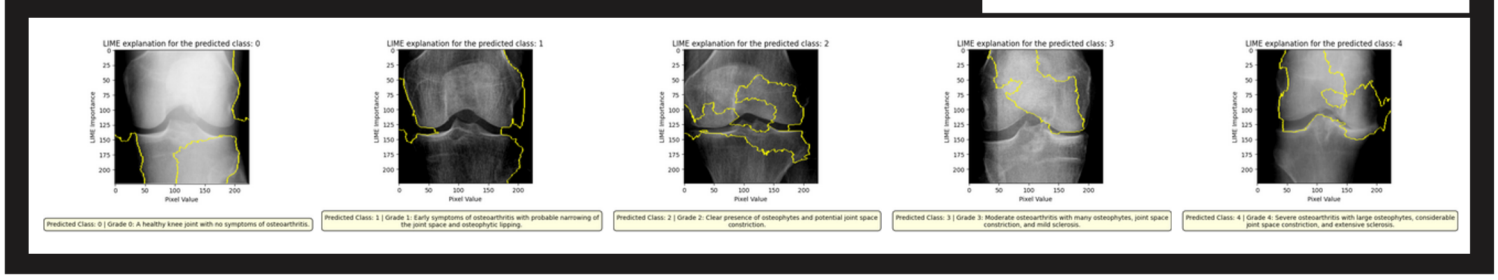
TABLE IX: Testing Report for Xception Model

Class	Precision	Recall	F1-Score
0	0.94	0.76	0.84
1	0.59	0.91	0.72
2	0.93	0.80	0.86
3	0.94	0.98	0.96
4	0.99	1.00	0.99
Accuracy		0.84	
Macro avg	0.88	0.89	0.87
Weighted avg	0.87	0.84	0.84

INNOVATIONS IN MODEL INTERPRETABILITY: A DUAL-METHOD APPROACH

SHAP and LIME provided complementary insights, with SHAP offering a global view of key features and LIME enabling case-specific interpretation. This integration validated feature importance, increasing trust in the model. SHAP revealed feature interactions, while LIME identified localized influences, addressing biases. Tailored for medical imaging, the approach ensured relevance to knee osteoarthritis grading and supports the ethical adoption of AI in healthcare.





CONCLUSION

In conclusion, my project on "Knee Osteoarthritis Detection and Severity Grading with SHAP and LIME Interpretability" successfully integrates machine learning with explainable AI to address a critical issue in medical imaging. By using deep learning models, we accurately classified knee osteoarthritis images into severity categories. The application of SHAP and LIME enhanced interpretability, providing both global and local insights into the model's decision-making process. This approach not only improves transparency but also builds trust in the model's predictions, making it valuable for clinical use in early detection and grading of knee osteoarthritis.